Learning How Wrong Offline Estimates Are

Reliable Launch Decisions for Recommender Systems

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The problem

- A shopping app wants to replace its recommender (the model that picks which products to show).
- Testing on real users (A/B test) is slow and risky.
- So it first **estimates offline**, from old click logs, how the new recommender would have done.
- Estimates have errors, so it launches only if

$$\text{estimated improvement} - \text{safety margin} > 0$$

Question

Is the usual safety margin big enough? **On real data: no.** It is about half as big as it should be.

The idea in one picture: the friend who is always late

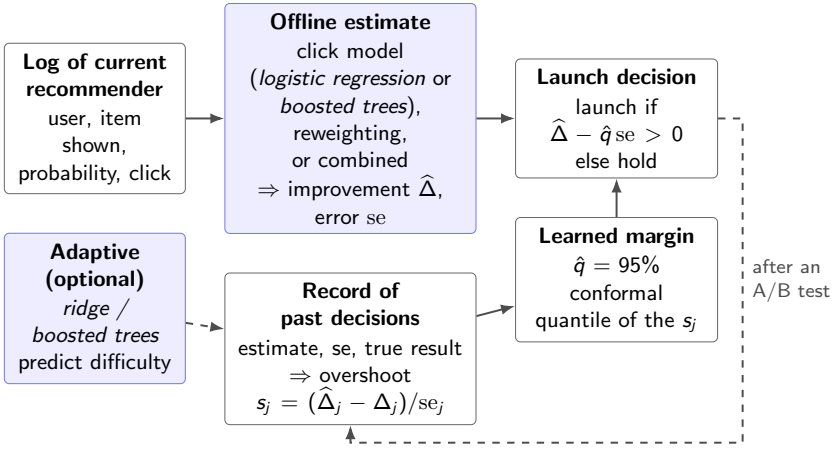
- Your friend says "10 minutes".
- Looking at the last 20 times, they arrived up to **25 minutes** late.
- So you plan for 25 minutes. You do not need to know *why* they are late.

Our method does the same for estimates:

- Look at past decisions where we know both the offline estimate and the real result.
- Measure how wrong the estimates actually were.
- Set the safety margin from that track record.

Conformal prediction is the statistically correct way to pick the number, so that it is right 95% of the time.

The method at a glance



Blue boxes contain the machine-learning models. The dashed arrow: every A/B test adds one more past decision, so the margin keeps learning.

Step 1: estimating a new recommender offline

Log of the current recommender: user x, item shown a, probability it was shown p, click r.

1. **Click model**: train a classifier for $P(\text{click} \mid \text{user}, \text{item})$ (**logistic regression** or **gradient-boosted trees**); average its predictions over what the new recommender would show.
2. **Reweighting**: count logged clicks, weighting items the new recommender likes more: $w = \dfrac{P_{\text{new}}(\text{item})}{P_{\text{old}}(\text{item})}$, estimate = average of $w \cdot r$.
3. **Combined**: click model, plus reweighting of its mistakes.

Improvement estimate $\widehat{\Delta}$ = new click rate − old click rate, with standard error se.
All written from scratch; match the `obp` reference library to 10^{-9}.

Step 2: the learned margin

Usual rule: launch if $\widehat{\Delta} - 1.645\,\mathrm{se} > 0$.

Our rule:

1. For each past decision j: how many standard errors did the estimate overshoot?

$$s_j = \frac{\widehat{\Delta}_j - \Delta_j^{\text{true}}}{\mathrm{se}_j}$$

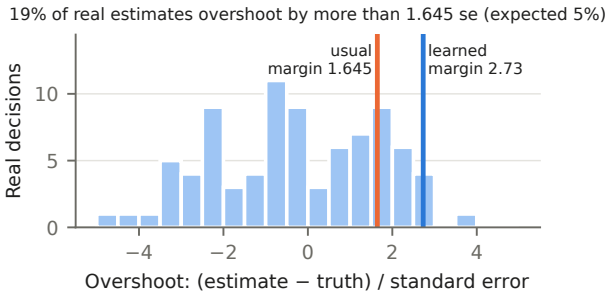
2. Sort them; take $\hat{q}$ = the value that 95% of them are below.
3. Launch if $\widehat{\Delta} - \hat{q}\,\mathrm{se} > 0$.

If estimates were honest, $\hat{q} \approx 1.645$. On real data, $\hat{q} \approx 2.7$.

Guarantee

If the new decision is like the past ones, the margin is big enough at least 95% of the time, *even if the estimator is biased*.

Why 1.645 is too small: the real record



- Each bar: real decisions, by how many standard errors the estimate overshot the truth.
- Usual margin assumes only 5% beyond 1.645.
- Real: **19%** beyond it.
- Learned margin **2.73**: 5% beyond.

Why the guarantee holds

- The margin is big enough exactly when $s_{\text{new}} \leq \hat{q}$.
- If the new decision is like the past ones, s_{new} is equally likely to land anywhere in the sorted list of $m + 1$ scores.
- We picked $\hat{q}$ so that at most 5% of positions are above it.
- So the margin fails at most 5% of the time.

Assumes: new decisions resemble past ones. **Does not assume:** unbiased estimator, correct click model, normal errors.

Data

Real: Open Bandit Dataset

- 26M recommendations, fashion e-commerce site, 7 days, 3 campaigns
- Two recommenders ran *at the same time* on split traffic: one random, one adaptive
- So each one's real click rate is the **true answer** when estimating it from the other's log
- 12-hour windows $\Rightarrow$ **84 real decisions**

Controlled: 6 classification datasets

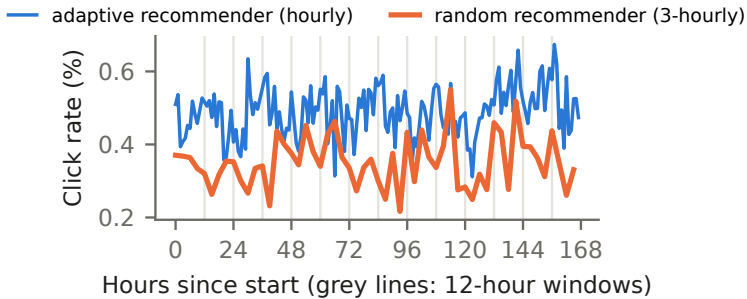
- Classes = items; click if the right class is shown
- True click rate of every recommender is exact
- New recommenders: retrained, naive Bayes, k-NN, random forest, ...
- Population shift created with **PCA**
- **2,688 decisions**, 27% not better

Finding 1: real estimates are about 2× more wrong than they say

	Actual error / reported error (1.0 = honest)		
Estimator	usual se	grouped by page	**grouped by hour**
Click model (GBT)	2.62	2.41	1.99
Reweighting	1.67	1.62	1.22
Combined (GBT)	1.80	1.76	1.13

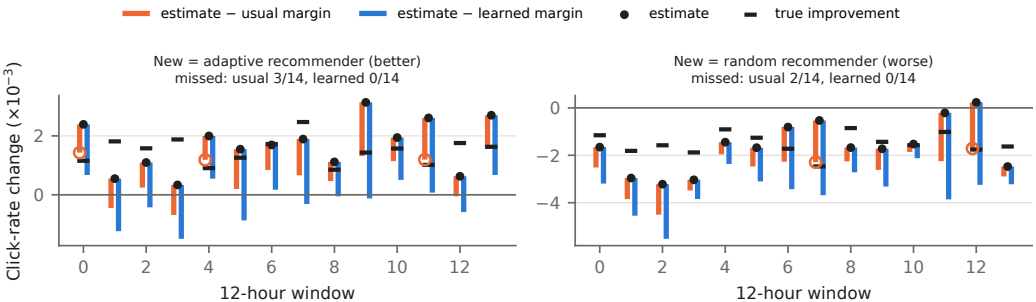
- The usual 95% margin is big enough only **60–80%** of the time.
- Grouping the 3 items on one page: no change ⇒ not the cause.
- Grouping by hour: explains most of it ⇒ **user behaviour drifts hour to hour**.
- Click models stay at ∼2: their error is **bias** from an imperfect classifier.

Why: user behaviour drifts hour to hour



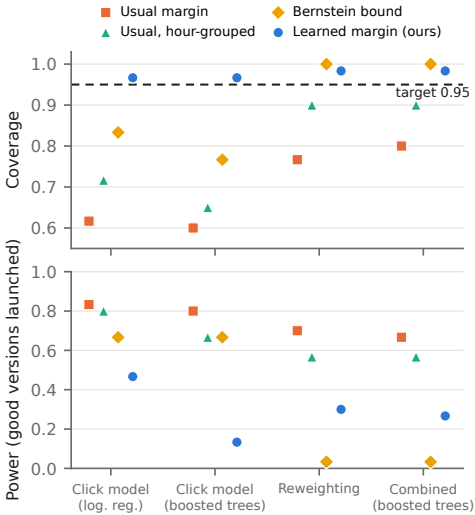
- Click rate changes within each 12-hour window (grey lines).
- The usual standard error treats every row as independent, so it misses this.
- Grouping errors by hour explains most of the gap, but not the click model's bias.

Finding 2: every real decision, one campaign



Bars: estimate down to estimate − margin. Launch if the bar's bottom is above 0. Circle: the truth fell below the bar (a miss). All three campaigns: usual margin misses **16/84**, learned margin **4/84**.

Finding 2: the learned margin is safe on real data

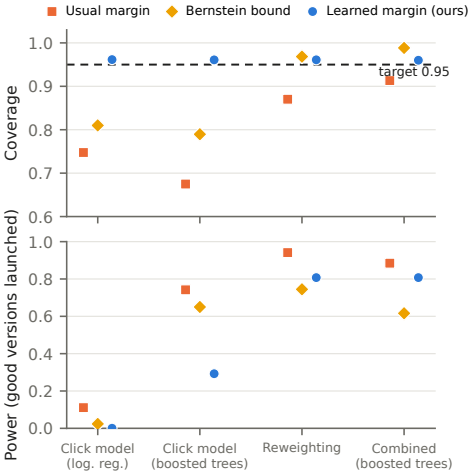


Each window uses only *earlier* windows as history.

- Usual margin: safe 60–80%
- Usual, hour-grouped: 65–90%
- **Learned margin: 95–98%**
- Bernstein bound: safe, but almost never launches

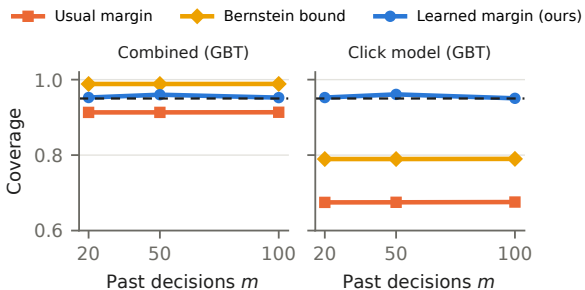
Cost: fewer launches (13–47% vs 63–83%). That is the price of honest uncertainty.

Finding 3: controlled experiments with exact truth



- Usual margin: safe 68–91%
- **Learned margin: 95–96%** for all 10 estimator variants
- Biased click model: wrong launches **27.7%** → **0%**
- History from other datasets: still 93–95%
- Breaks under deliberate population shift (78–88%), as the assumption predicts

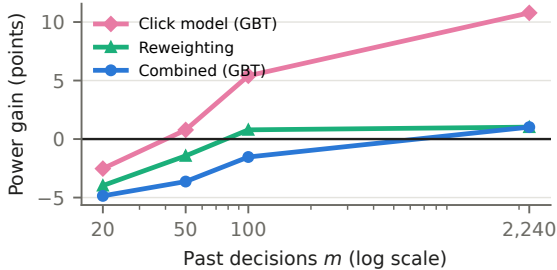
How much history is needed?



- The learned margin hits the 95% target with only **20** past decisions.
- More history does not change coverage; it makes the margin more stable.
- The usual margin stays below target however much data you have.

Adaptive margin: an ML model learns which decisions are hard

- Inputs: log size, largest/average weight, weight spread, $\log \mathrm{se}$
- **Ridge regression** or **gradient-boosted trees** predict how wrong the estimate will be
- Hard decisions get a bigger margin, easy ones a smaller one



Helps only with lots of history. With few past decisions, the plain version is better.

Where the machine learning is

Model	Used for
Logistic regression	Click model; the current recommender in controlled experiments; retrained versions
Gradient-boosted trees	Click model (stronger); difficulty model in the adaptive margin
Ridge regression	Difficulty model in the adaptive margin
Naive Bayes, k-NN, random forest	New recommenders in controlled experiments
PCA	Creates the population shift
Cross-fitting	Models never predict on their own training rows
Conformal prediction	Turns the track record into a guaranteed margin

No deep learning. No reinforcement learning: nothing learns by trial and error; we only analyse existing logs.

Live demo: the one window where the new recommender was worse

```
$ make demo
Estimated improvement in click rate: +0.00307 (standard error 0.00116)

Rule                                          est - margin  Decision
Usual margin (1.645 x se)                         +0.00116  LAUNCH
Usual margin, hour-grouped se                     +0.00054  LAUNCH
Bootstrap margin                                  +0.00120  LAUNCH
Bernstein bound                                   -0.00378  hold
Learned margin (2.73 x se, 72 past decisions)     -0.00010  hold

True improvement, measured on real traffic: -0.00035
  -> new recommender is NOT better
```

The usual margins launch a worse recommender. The learned margin holds.

Conclusion and limitations

- Offline launch decisions are only as safe as their margin.
- On real logs the usual margin was about **2× too small** (hourly drift + click-model bias).
- Learning the margin from past decisions gave the promised 95% safety on real and controlled data.

Limitations: needs a history of past decisions; assumes the future resembles the past; fewer launches; real data has only two recommenders.

Next: weight past decisions by similarity; combine with hour-grouped standard errors.

Backup: all main results

	Coverage				Wrong-launch rate				Power			
Estimator	Usual	Hour	Bern.	Ours	Usual	Hour	Bern.	Ours	Usual	Hour	Bern.	Ours
Real data (60 decisions, history = earlier windows)												
Click model (LR)	.617	.717	.833	**.967**	.033	.033	.033	.033	.833	.800	.667	.467
Click model (GBT)	.600	.650	.767	**.967**	.033	.033	.033	.000	.800	.667	.667	.133
Reweighting	.767	.900	**1.00**	**.983**	.033	.033	.000	.033	.700	.567	.033	.300
Combined (GBT)	.800	.900	**1.00**	**.983**	.033	.033	.000	.000	.667	.567	.033	.267
Controlled experiments (exact truth, $m = 50$ past decisions)												
Click model (LR)	.748	–	.810	**.962**	.013	–	.000	.000	.111	–	.024	.001
Click model (GBT)	.675	–	.790	**.961**	.277	–	.215	.000	.742	–	.650	.293
Reweighting	.870	–	**.968**	**.961**	.000	–	.000	.000	.941	–	.745	.807
Combined (GBT)	.913	–	**.988**	**.960**	.005	–	.000	.001	.885	–	.616	.808

Coverage: margin big enough (target 0.95, bold if met). Wrong-launch rate: not-better recommenders that were launched. Power: better recommenders that were launched. Usual = 1.645 se; Hour = hour-grouped se; Bern. = Bernstein bound; Ours = learned margin.

Backup: adaptive margin, ridge vs boosted trees

Estimator	Model	m=20	m=50	m=100	Large	Real
Click model (GBT)	Ridge	−5.2	−0.3	+1.7	+9.0	+10.0
	GBT	−2.5	+0.8	+5.4	+10.8	−6.7
Reweighting	Ridge	−4.8	+0.9	+1.4	+1.0	−16.7
	GBT	−4.0	−1.4	+0.8	+1.0	−13.3
Combined (GBT)	Ridge	−6.8	−1.2	−0.6	+0.9	−3.3
	GBT	−4.9	−3.6	−1.5	+1.0	−13.3

Change in power (percentage points) of the adaptive margin compared with the plain learned margin. m = past decisions; Large = history from five other datasets ($m \approx 2{,}240$); Real = real-data history (24–78). Coverage stays within 0.91–1.00. Both difficulty models gain only with a large history.

Backup: what is new compared with prior work

Prior work	Difference
Choosing the best estimator (2023–24)	We make the *decision* safe for any estimator
High-confidence bounds (2015)	Valid only if the estimator is unbiased
Conformal intervals for single outcomes (2022)	We calibrate launch decisions using past decisions
Offline A/B testing (2018)	Uses only within-log variability